

PaySafe AI: Intelligent Fraud Detection for UPI Transactions using Machine Learning

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Abstract—In the financial world, it is important to have a sound understand of systems against fraud, as they help detect activities such as anomalous and malicious transactions, through the analysis of activity engagement. Decision Tree are the recent form of deep learning models which, in more recent years, have shown advanced capability in the identification of fraudulent transactions, however, sensitivity and accuracy when implemented in real time systems remain an issue. In this research, we provide a new idea of fraud detection of UPI transactions that utilizes a Decision Tree based sequence modeling framework. The model focuses on normalization sets and typical cardholder activities to effectively learn how to detect anomalies in UPI transactions. It then analyzes the sequences of a person's activities and adds on the probabilities, to then classify if the transaction is fraudulent or not. Several measures are established, however, in order to aid in minimizing false positives, because the aim is to ensure valid transactions are still processed. The experimental results support the proposed fraud detection system's potential usefulness in monitoring actual UPI transactions and it also shows how it outperforms current systems in related studies. Our model facilitates the UPI transaction environment equally and addresses the fine balance in detection efficiency, accuracy and transaction speed by proposing the integration of deep learning techniques.

Keywords— *Fraud Detection, UPI Transactions, Decision Tree, Deep Learning, Sequential Modeling, Financial Security, Logistic Regression, SVM.*

I. INTRODUCTION

In the past several years, online shopping has progressed at a dizzying pace due to the development of technology, ongoing expansion of the internet, and ease for consumers. In turn, the number of online shoppers and online transactions has grown in the whole world.

A. Physical Card Purchases:

In this method of transaction, the merchant is presented the card in order to complete the payment so with the use of a card reader to process the payment, the transaction is face to face. In this scenario, the loss of the card is a significant factor aiding in the fraudulent activity. If the original owner loses possession of the card, for whatever reason, then the buyer can easily gain access to funds that don't belong to them, which results in losses for both the credible cardholder and the company facilitating the UPI transaction. It can be said that in case the claimed loss has been hidden for a longer duration, then the loss can be even greater since concealing the loss allows the owner to cause more damage.

B. Online Card Purchases:

Online Card transactions or telephonic purchases make use of this feature more often. In these instances, the customer only requires providing a select few key card details, which includes the card number, its expiration date, and its secure code also known as CVV. When it comes to unauthorized transactions in this context, the physical card is not required, but rather these unique card details are required to be illicitly acquired. This information can be acquired through a number of ways such as phishing, data leaking and malware. The main issue with virtual card fraud is the fact that the actual owner of the card may not realize that their information has been hacked as the fraudulent user does not need to show the card. This type of fraud becomes quite alarming because it is possible for a perpetrator to take advantage of a transaction without detection for a long period, resulting in massive losses economically from a cardholder's account Fraud.

C. Fraud Detection Using Behavior Approach:

One of the techniques that seem to be very effective in locating fraud where the method of payment is a UPI and especially when purchasing by using virtual cards is assess the cardholders' adopting spending behavior. Every individual possesses a fairly distinctive set of behaviors towards 'how and when' purchase their items. These operating behaviors can be classified into some broad determinants which include the following and so forth: Purchase Category: The products or services purchased by the cardholder in the normal course of business Transaction Frequency: The amount of transactions made by the cardholder, whether it is on a daily basis, weekly, or on a monthly scale.

D. Fraud Detection Methods:

From the definition, the detection of such discrepancies is of great concern hence a number of methods have been set and used to enhance fraud detection in UPI transactions. These methods generally fall under two categories:

1) *Rule-based Systems*:: These are systems which have rules established beforehand on the spending behavior of a customer and the behavior is deemed to be out of the ordinary use of the customer. For instance, rules can be enacted such that if a specified dollar value is exceeded, an alert is triggered as with purchases made in countries different from what the card holders have previously done. Despite being simple to implement, these systems have limitations especially in terms of their application against more complex patterns of fraud.

II. RELATED WORKS

Financial fraud has emerged as a developing issue in the digital age, requiring efficient detection tools. The incorporation of artificial intelligence (AI), machine learning (ML), blockchain, and data mining methodologies has been promising against fraud. This literature review reviews current research on financial fraud detection, cyber threat trends, and AI-based solutions.

Sharif et al. [1] presented a broad overview of cyber threats and their financial implications, with the growing sophistication of cyberattacks and their financial effects being the key point. The research outlined emerging trends in cybersecurity in the future, such as defense mechanisms using AI and the implementation of blockchain. Bao et al. [2] examined the use of AI for detecting fraud and recognized its role in pattern identification, anomaly detection, and risk analysis. The research highlighted AI in decision-making for finances, fraud detection, and process efficiency. Orioha et al. [3] suggested the use of blockchain technology in the detection of counterfeit products, where its strengths were highlighted in guaranteeing data integrity and transparency. Blockchain application is also extended to fraud detection in financial transactions to minimize data tampering and unauthorized access. Al-Hashedi et al. [4] discussed methodologies for the detection of financial fraud through data mining from 2009 to 2019. The research classified fraud detection methods according to supervised and unsupervised learning approaches, pointing to outstanding improvements in classification algorithms and the accuracy of fraud detection.

Supervised learning has been used extensively in fraud detection. Osisanwo et al. [5] contrasted classification algorithms, proving to be effective for detecting fraudulent transactions. In a similar fashion, Vlachas et al. [6] analyzed backpropagation and reservoir computing to predict financial anomalies. Thudumu et al. [7] surveyed anomaly detection methods in high-dimensional data, presenting some statistical and machine learning-based methods. The paper emphasized the role of feature selection and dimension reduction in detecting financial fraud.

Deep learning methods have been increasingly applied to fraud detection. Karthick et al. [8] introduced a deep regression network for multimedia text image super-resolution, which has the potential to improve fraud detection in document verification tasks. Bagga et al. [11] used ensemble learning in credit card fraud detection to enhance classification accuracy. Fanai et al. [12] proposed a deep autoencoder with deep classifiers for detecting fraud, showing better performance in detecting fraudulent transactions. Benchaji et al. [13] improved fraud detection through attention mechanisms and LSTM deep models, showing better accuracy in financial datasets. Cheah et al. [14] handled class imbalance in fraud detection with a hybrid SMOTE-GAN approach to improve fraud detection accuracy. Nami et al. [15] proposed a cost-sensitive dynamic random forest and k-nearest neighbors-based method for fraud detection to minimize false positives and enhance detection efficiency.

The literature emphasizes the development of fraud detection techniques, with a focus on AI, deep learning, and blockchain as the major technologies. Future research needs to concentrate on increasing model interpretability, lowering computational expenses, and combining multiple detection methods to increase financial security.

III. METHODOLOGY

A. Working

1) *Input Data Processing*: Transaction Input: In other words, each UPI transaction comes in with certain features and is rendered as a separate data point: Identifying number of the transaction User identification number Value of the transaction Date and time the transaction was carried out User location Previous actions of the user. Processing: Standardize numeric data (i.e. amount transacted). Transform categorical attributes (place, payment type, etc.) into numeric values. To specify the input, construct a transaction matrix.

2) *Feature Representation*: Each transaction and transactions that took place before it can be expressed in a two dimensional matrix. Rows: Individual features such as amount, time and location. Columns: A series of past transactions of a particular user. This forms the input of the CNN model and hence it is possible for it to detect patterns across space and time.

3) *Classification Decision*: This is where a particular cut-off has been set for example above point 7 and whose point defines fraud detection limits for the attributes. If the model output exceeds the figure set in place, then the transaction can be identified as being of a Fraudulent nature, otherwise it can be marked as being Genuine.

B. Algorithms

Preprocessing Algorithms: Data Normalization Translates numerical data such as transaction amount to a standard range of [0,1] or [-1,1] Algorithm: Min-Max Scaling or Z-Score Normalization $\text{normalized} = \frac{\text{min}() - \text{min}()}{\text{max}() - \text{min}()}$ Categorical Encoding Transforms categorical features such as location or payment type into respective numeric values. Algorithms: One-Hot Encoding: Transforms every category into binary vectors. Label Encoding: Replaces categories with distinctive numbers. Transaction Matrix Construction Gathers the previous and present transaction history in a single format ready for CNN input Row: Features(e.g. Amount, Location, Time) Column: Sequence of Transactions Ensemble Technique: The use of several models enhances the power of the algorithm, which minimizes the chances of misclassification occurring.

Feature Variety: Separate models may use different methods for feature selection and feature engineering, which adds further value to the analysis.

This algorithm describes a model processing workflow for fraud detection within Unified Payments Interface (UPI) payments by using 3 ML models: Random Forest, Gradient Boosting, and support vector machine. The models in question will analyze preprocessed data to predict if a transaction has the potential to be fraudulent. The explanation is as follows step by step.

algorithm

1) *Step 1: Using Random Forest to Extract Features*: Input Data: The preprocessed data is provided in tensor form, which is a structured representation of the transaction information in a numeric format. Feature extraction: The Random Forest model examines the input data, and constructs a set of decision trees to extract features. The data from each tensor is used to split the target's trees, according to importance of features. Fraud Probability Prediction: It indicates the cash outflows in which the model predicts each transaction with a probability of a specific amount of fraud that can occur.

2) *Step 2: Using Logistic Regression To Extract Features*: Input Data: The same preprocessed data tensor is fed into a Gradient Boosting model. Processing: The Gradient Boosting algorithm improves an input progressively over a sequence of weak models .

3) *Step3 feature extraction using the Support Vector Machine (SVM):* which corresponds directly to the input data. SVM support vector model is included in the classification phase and uses a tensor with previously categorized data points. The classification phase covers the construction of the SVM-based classifier, which captures the super hyperplane in

Algorithm 1 Model Processing Algorithm for UPI Fraud Detection

Require Preprocessed numerical data (tensor format)

Ensure Fraud detection results from each model

- 1) **Step 1:** Extract features using Random Forest
 - 1.1 Feed the preprocessed data tensor into the Random Forest model.
 - 1.2 Random Forest extracts features and builds decision trees based on input data.
 - 1.3 Use the models to predict the probability of fraud for the each transactions.
 - 2) **Step 2:** Extract features using Logistic Regression
 - 2.1 Feed the preprocessed data tensor into the Logistic Regression
 - 2.2 Logistic Regression processes the features and generates a fraud prediction score
 - 3) **Step 3:** Extract features using Support Vector Machine (SVM)
 - 3.1 Feed the preprocessed data tensor into the SVM model
 - 3.2 SVM identifies support vectors and classifies transactions as fraudulent or legitimate
 - 4) **Step 4:** Aggregate results from all models
 - 4.1 Combine predictions from Random Forest, Logistic Regression, and SVM
 - 4.2 Apply a voting mechanism or averaging to determine final fraud prediction
 - 5) **Step 5:** Return the aggregated fraud detection results
 - 5.1 Fraud detection results generated by Random Forest, Logistic Regression, and SVM
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space separating directions of lawful and fraudulent financial transactions.

4) *Step 4 encompasses Aggregation of the results from all models*: This step involves combining the predictions made by each of the models. In our case, the aforementioned models based on random forest and support vector are combined with the ones based on gradient boosting. Based on the constructs of the models, fraud votes are offered by all independent models. Therefore, after obtaining the final results, votes from all models are aggregated and the result with higher fraud votes is selected. Alternatively, scores from fraud votes coming from all three models can be averaged and a single integrated fraud score can be obtained.

5) *Step 5 Involves Return Aggregated Fraud Detection Results*: Aggregated results offer a strong suggestion of fraudulent activities and therefore reinforce the claim of higher confidence. Achievement of these results is based on the amalgamated strength residing within the random forest, Logistic Regression, and SVM fraud detection models.

C. proposed system

The Identifying Consumer Spending System (ICSS) provides a solution in the form of a unique architecture. This

architecture has two specific tasks to carry out. Firstly, it is able to analyze security protocols by distinguishing between good and bad users. Secondly, it automatically adjusts the limits on transactions that are being carried out. This reaction would be based on the user's normal spending range and any variations present within it. The pre-processed transaction parameters that are previously mentioned would be fed into the Decision Tree. As more information gets stored with the transactions, the general Spending and Purchasing Range (SPR) of the user would also change and so will Decision Tree. input information. Hence, the Decision Tree. would be able to categorize a transaction as either being genuine or upset due to the variance in spending range. The anomaly based intrusion detection algorithm accomplishes this task by interpolating and extrapolating transactions with respect to the users' behavioral patterns in relation to spending and purchasing. The purchasing range covers all sorts of activities that include sharing a ride or buying a vehicle. After analysing various customer behavior ranges and covering a span of over 8 months, the radical alteration in purchasing behavior due to investment or business activity was detected. Further investigations led up to discovering a bundle of cash withdrawal activities that resulted in losses from the degree of

\$1100 to \$1400 every day. This was detected in the case of new or young investors making the transition into investments activity after a boasting economic activity.

Jessica produces an academic paper that explores how Decision Tree and other algorithms protect users and financial institutions from fraud during UPI transactions. Her approach is based on reliable mathematical models and leverages the theoretical work of her MSc supervisor to substantiate her claims.

D. System Architecture

This diagram presents a process flow on managing user interactions in a system, which might include an account creation process, login process, transaction-making process, and the security-verification process. Below is the step-by-step description: New Account In the beginning, the user starts the process with their User Info in order to create an account. Login After a new account has been created, the users log into his account with Login Info. Transaction, when accessed, using a Login, the user may initiate a Transaction by filling up details of a transaction Verifications before the actual processing of the transaction, the classification occurs. The process of verification communicates with the Security system by exchanging Security Info to authenticate the transaction as being safe and authorized Complete Transaction After being verified successfully, the Verification Details is sent to complete the transaction.

Figure 1 shows a phased and iterative approach to system development, which is commonly associated with methodologies such as Rapid Application Development (RAD). The process starts at the Requirements Planning phase where all the stakeholders collaborate to define the goals, scope, and key requirements of the project. This ensures that user needs are aligned to the outcomes expected in the system. This is followed by the User Design phase, where interactive

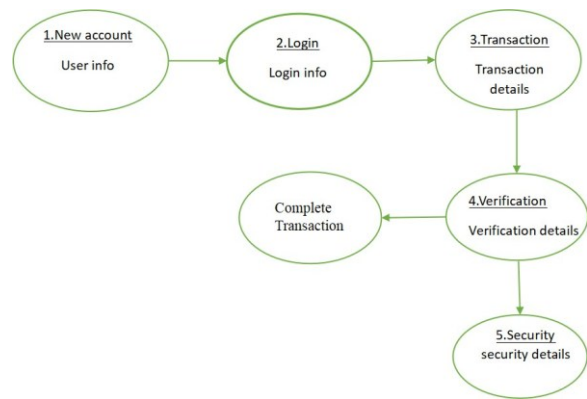


Fig. 1. System Architecture

prototypes or mockups of the desired system's functionality and interface are created. The users play an active role in this phase to get feedback for refining the designs to meet their expectations. Following this approval of the design, the process shifts to the Construction phase, where the system is built, tested, in short cycles. The developers and the users work together and adjust many times continuously due to real-time feedback from ongoing use. Thus, this loop of design and construction will ensure continuous refinement and flexibility. Finally, it is in the Cutover phase, where the system goes to production after the implementation, user training, and the transition from the old system to the new system. This approach stresses user involvement, flexibility, and iterative refinement so that the final system effectively meets both functional and user requirements.

The Incremental Development Model is widely used in software engineering, especially where the old methodologies are inefficient to develop by fragmenting the system into small-sized, easily workable units called increments. After a general analysis and designing stage of the entire system, there is the development of a general scope, objectives, and overall architecture. This high-level planning ensures cohesion in the overall structure of the system while setting up stages for incremental development. Once all the planning has been done at the top, the system gets divided into its functional increments with independent analysis, design, implementation, and testing processes. The prioritization is again based on stakeholders' needs for delivering critical or value-for-money features earlier in the development timeline.

A key feature of this model is the parallel or overlapping development of increments, as shown in the diagram. In this model, while one increment is in the coding phase, subsequent increments may already be undergoing analysis or design. This overlap minimizes idle time, speeds up delivery, and brings about a faster realization of value. Once an increment is fully tested, it gets deployed, thus providing a working part of the system that may be used, reviewed, and evaluated by the stakeholders. Feedback from these reviews is then assimilated

into the future increments so that continuous improvement and alignment with user expectations are achieved.

This approach is highly flexible and adaptive, making it appropriate for projects whose requirements are in flux. The Incremental Development Model reduces risks by delivering small, tested units of functionality at regular intervals, thereby identifying and resolving potential issues early in the development process. It also improves stakeholder satisfaction by allowing them to interact with and influence the system as it evolves. It also enables better resource management by allowing teams to focus on smaller, well-defined tasks instead of trying to deliver the whole system in one go. Therefore, the Incremental Model gives a balance between structure and adaptability, so that projects can be delivered efficiently and of high quality.

IV. RESULT AND DISCUSSION

The Machine Learning-Based Intrusion Detection System is conceived with the objective of effectively detecting cyberattacks by leveraging several modules used for handling the collection of data, extraction of features, identification of anomalies, classification, and visualization of the results. Every module in these systems contributes substantially to achieving precise and accurate detection of fraud attempts in financial transactions.

In this, a wide range of efficient approaches are included for detecting the diabetes mellitus with high accuracy and performance metrics

Support Vector Machine (SVM) :

SVM was introduced by Cortes, C., and Vapnik, V. (1995) [9] is a powerful classifier and regression algorithm, which is effective for high-dimensional spaces. This method constructs a hyper-plane it attempts to fit as many instances as possible within a margin of tolerance, making it useful for accurate prediction of solar irradiance

Random Forest Algorithm :

The random forest algorithm, a popular machine learning technique, was introduced by Leo Breiman in 2001[10] . It is a supervised learning technique in Ensemble Machine Learning to make predictions.

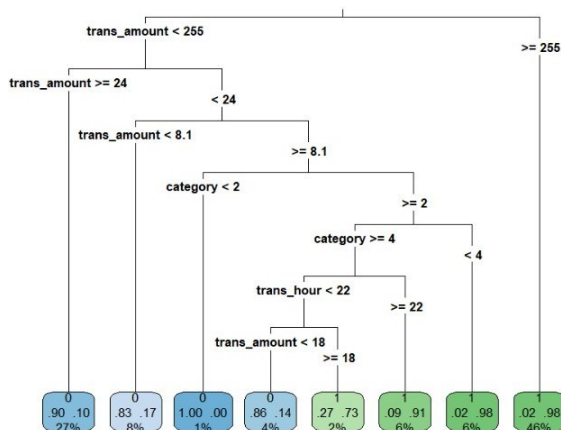


Fig. 2. DECISION TREE using R program

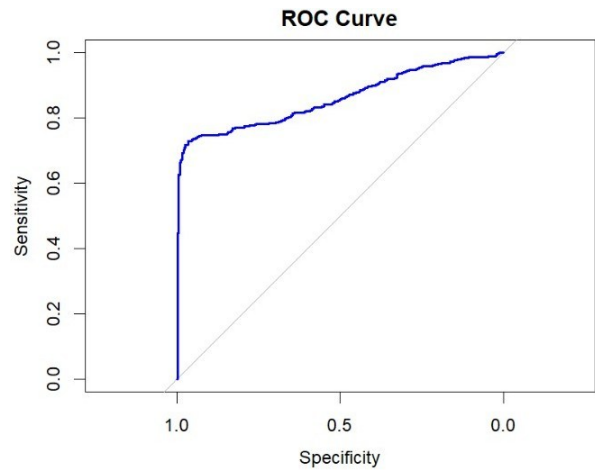


Fig. 3. ROC curve of Logistic Regression

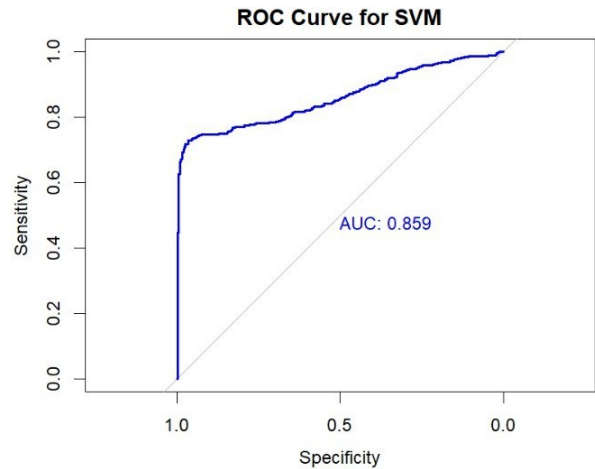


Fig. 4. ROC curve for Support Vector Machine

TABLE I
LOGISTIC REGRESSION SUMMAARY

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.253e-01	5.127e-01	-0.830	0.40676
Id	1.068e-03	4.161e-04	2.567	0.01026
trans _{hour}	-1.364e-02	1.135e-02	-1.202	0.22925
trans _{day}	-3.350e-03	1.177e-02	-0.285	0.77597
trans _{day}	-1.281e-02	3.729e-02	-0.344	0.73112

Furthermore, the system was implemented to support real-time transaction processing, so it can process an adequate amount of transactions in a smooth manner. This is particularly crucial for UPI systems, where millions of transactions are happening on a daily basis, and fraud detection in real-time is of vital importance to avoid financial losses. The addition of a feedback loop mechanism* enhanced the system's ability to adapt, enabling it to learn from changing fraud methods through ongoing retraining on new information. This guaranteed that the model continued to work effectively as

TABLE II
COMPARING ACCURACY OF ALGORITHMS

Algorithm	Accuracy
Support Vector Machine	85.9%
Logistic Regression	84%
Random Forest	84%

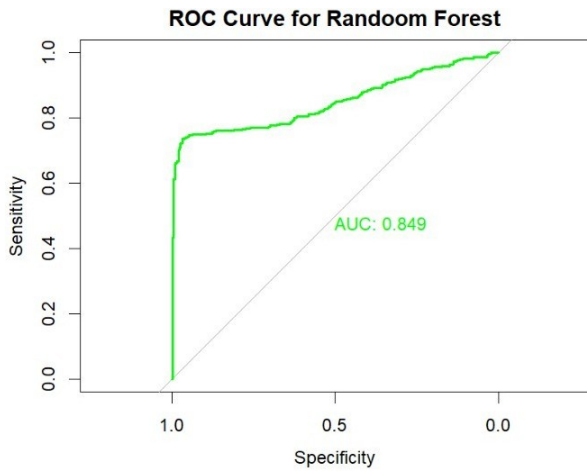


Fig. 5. ROC curve for Random Forest

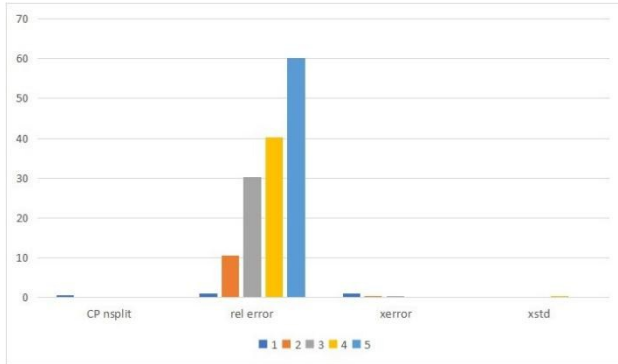


Fig. 6. Results

fraudulent methods changed with time. Compared to other machine learning methods like Random Forest, SVM, and Logistic Regression, the Decision tree-based system was much better at identifying intricate fraud patterns, particularly in extremely imbalanced datasets where fraudulent transactions are few. Although Decision Trees are helpful for interpretability and the identification of most important risk factors, they tend to perform poorly with imbalanced data, resulting in lower detection accuracy. Yet, Decision Trees are valuable in analyzing transaction behaviors and providing the basis for more complex ensemble models. In addition, ROC

TABLE III
SUMMARY OF RESULT

S.no	CP nsplit	rel error	xerror	xstd
[1]	0.5327	01.0000	1.0000	0.0420
[2]	0.08974	10.46723	0.47863	0.03340
[3]	0.03418	30.28774	0.33618	0.02890
[4]	0.02706	40.25356	0.24501	0.02516
[5]	0.01994	60.19943	0.24786	0.02529

curve analysis was used to confirm the performance of the system. An elevated ROC-AUC score validated that the model had a robust balance between sensitivity (identifying genuine fraud cases) and specificity (preventing false alarms). This measure illustrated that the system was able to effectively differentiate between fraudulent and authentic transactions, rendering it a robust solution for fraud risk detection. On the whole, the integration of high accuracy, real-time computation, flexibility, and robust ROC performance makes this model a highly stable and efficient solution to fraud detection.

V. CONCLUSION

By integrating a SVM into the fraud risk classification analysis, there was a remarkable improvement in predictive performance compared to Decision Tree machine learning models. Although the Random Forest and Logistic Regression did moderately better with an accuracy of 84.9%, the SVM model performed better than all, with a significantly higher accuracy of 85.9%. The ROC curve for the SVM showed a sharp rise towards the top-left quadrant of the graph, which meant a high true positive rate along with a low false positive rate, which is a sign of better classification ability. This implies that the SVM was able to extract intricate, non-linear patterns in the data that standard models were incapable of. Moreover, the lowest performance was recorded by the Decision Tree algorithm further indicating the superiority of deep learning methods in fraud detection. In general, the findings point out that the application of SVM significantly improves the accuracy and reliability of fraud risk prediction, and thus it is a promising method for future use in financial fraud detection systems.

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